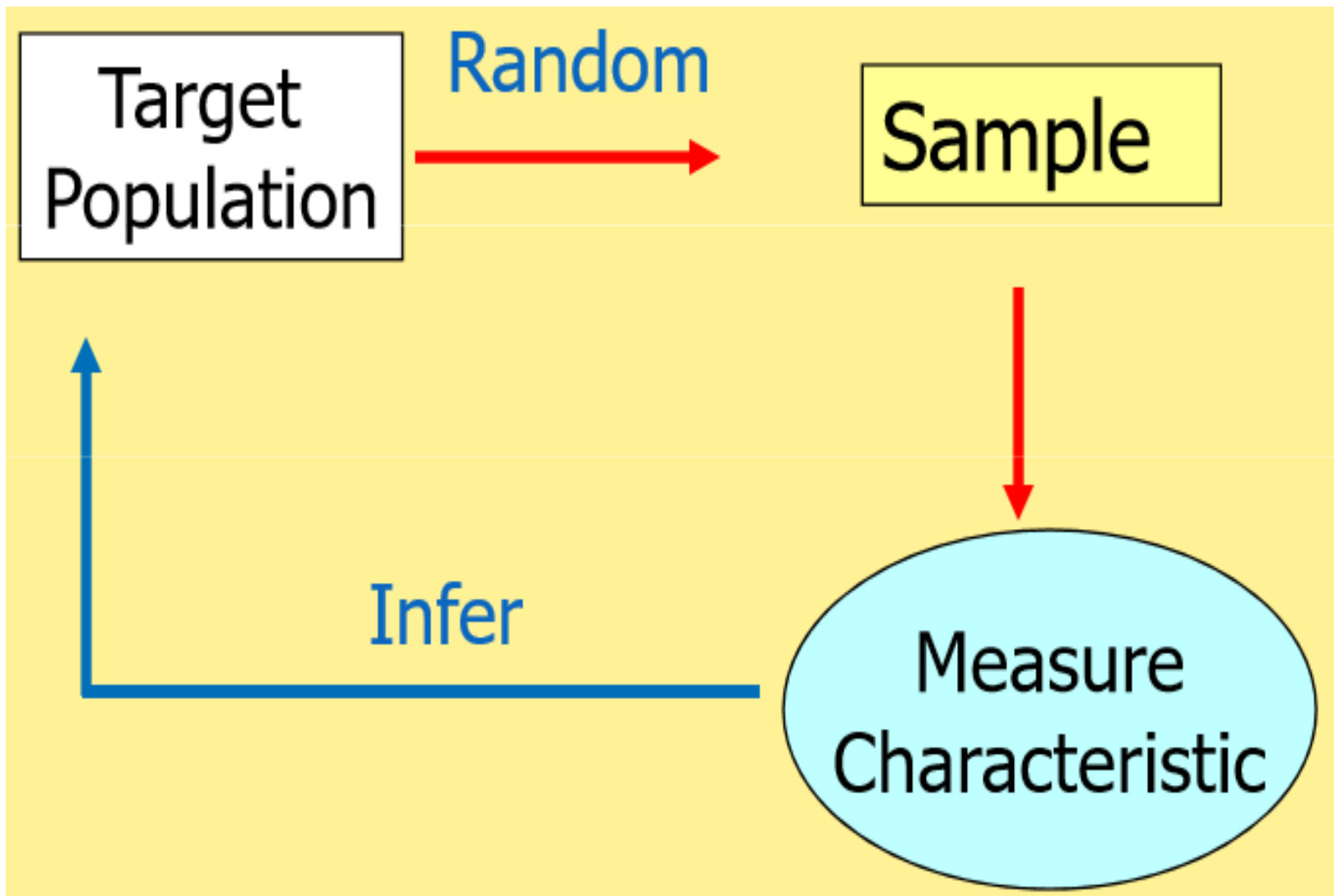


Hypothesis testing

Hypothesis testing

- The significance test or hypothesis test enables us to **measure the strength of the evidence** which the data supply concerning some proposition of interest
- Definition: A statistical hypothesis is an **assumption** or a **statement** which may or may not be true concerning one or more populations
- Eg. The mean height of class students is 1.63m
- There is no difference between the distribution of Pf and Pv malaria in Ethiopia (are distributed in equal proportion)

- Intr...



- **Types of Hypothesis**
- **There are two types of hypothesis**
 - **Null hypothesis (H_0)** – a statement claiming that there is **no difference** between population parameter and hypothesized value
 - **H_0 is always a statement about a parameter (mean, proportion, etc. of a population).**
 - **It is also called “**hypothesis of no difference**”**
 - **$H_0 : \mu = C$ Where**
 - **H_0 is the null hypothesis**
 - **μ is the population mean**
 - **C is the hypothesized mean**

- Cont... on type
- **Alternative hypothesis (HA)** – a statement claiming that **there is a difference** between population parameter and hypothesized value.
- HA must also be formulated before a sample is tested, so it, like the null hypothesis (Ho) **does not depend on sample values**.
- It is also called “**research hypothesis**” $HA : \mu \neq c$; ($HA : \mu > c$, $HA : \mu < c$)
- **Example:** the mean age of this class is 30 years
 $Ho : \mu = 30$ years $HA : \mu \neq 30$ years ($HA : \mu > 30$ years, $HA : \mu < 30$ years)

- **Errors in hypothesis testing**

- In hypothesis testing two specific types of errors can be committed
- **Types I error** – this occurs when rejecting of the null hypothesis while it was actually true.
- The probability associated to this error is α (level of significance)
- **Types II error** – this type of error occurs when accepting the null hypothesis while it was actually false
- The probability associated to this error is β ($1 - \alpha$, which is the level of confidence)

- **The two types of error can be summarized in the table.**

	Decision	
	Accept H_0 (Fail to reject H_0)	Reject H_0
H_0 true	Correct	Type I error
H_0 false	Type II error	Correct

- **Concepts of P – Values**

- **Another vital concept in statistical significance is the p-value, commonly reported in scientific journal.**
- **P-value is related to hypothesis testing,**
- **It is the probability of obtaining a result as extreme as (more extreme than) the one observed, if the null hypothesis is true.**
- **Is the probability of the observed outcome, assuming that chance alone was involved in creating the outcome i.e. assuming the null hypothesis correct, what is the probability that we would have seen the observed outcome.**

- **Conventionally p-value less than 0.05, the null hypothesis is rejected as unlikely**
- **It would lead to the rejection of the null hypothesis**
- **In this formulation, when we reject the null hypothesis we accept a complementary alternative hypothesis**
- **If $P > 0.05$ this is often taken as suggesting that insufficient information is available to discount the null hypothesis**

- **When P is below the cut off level (α), say 0.05, the result is called **statistically significant****
- **When above 0.05 it is called not significant**
- **It is important to distinguish between the significance level α and the p – value**
- **The significance level α is the probability of making a type I error. This is **set before the test is carried out****
- **The P – value is the result observed **after the study is completed** and is **based on the observed data****

- **Steps involved in testing a hypothesis**
 - **Step 1:** State the research question in terms of statistical hypothesis
 - The null hypothesis, H_0 is the statement claiming that there is no difference between the hypothesized value and the population value. (The effect of interest is zero)
 - The alternative hypothesis, H_A , is the statement that disagrees with the null hypothesis. (The effect of interest is not zero)
 - $H_0: \mu = 0$
 - $H_A: \mu \neq 0$

- **Step 2: Determine the level of significance (usually $\alpha=0.05$)**
- **Step 3: Statement of assumption**
 - Sample size large
 - Normal distribution
 - Variance is known
- **Step 4: In step 3 the sample size is large and variance is known use the following formula:**

$$Z = \frac{\bar{X} - \mu_0}{\sigma / \sqrt{n}}$$

- **If it is unknown variance and small sample size we use t-test statistics.**
- **T-tab depend up on the degree of freedom (i.e. n-1)**
$$t = \frac{\bar{X} - \mu_0}{S / \sqrt{n}}$$
- **For categorical data we use X²- test**
- **Step 5: Set the decision rule reject H₀ if Z-cal > Z-tab**
- **Step 6: Calculate the test statistics**
- **Step 7: Draw and state conclusion**

- **Hypothesis test about population mean (normally distributed)**
- **Known variance**
- **Example:**
 - Researchers are interested in the mean level of some enzyme in a certain population. They are asking: can we conclude that the mean enzyme level in this population is different from 25?

- **Step 1:** $H_0: \mu = 25$

$$H_A: \mu \neq 25$$

- **Step 2:**

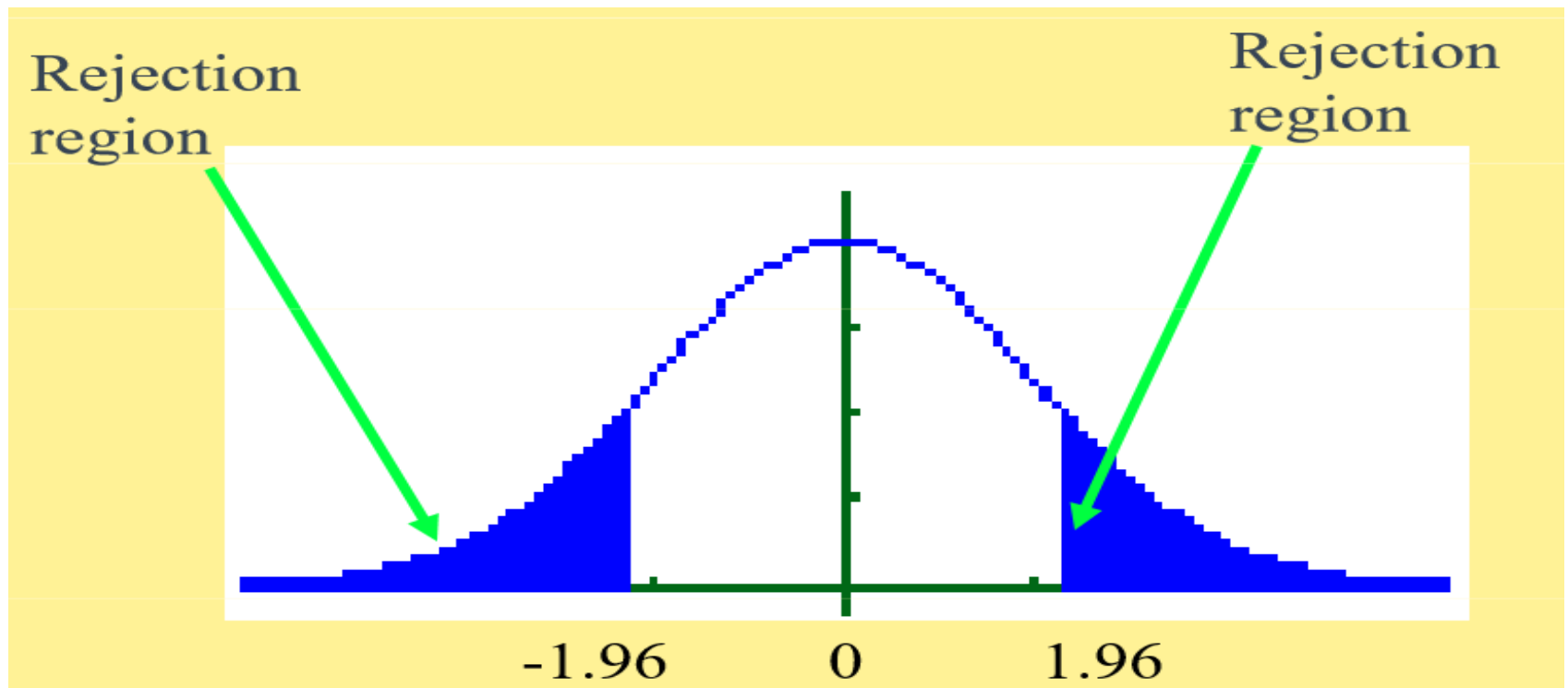
- They collect a sample of size 10 from a normally distributed population with a known variance, $\sigma^2 = 45$
- The calculated sample mean is = 22.

$$Z = \frac{\bar{X} - \mu_0}{\sigma / \sqrt{n}}$$

- **Step 3:**

- Testing a hypothesis about population mean
- The population is normally distributed
- Population variance is known $\Rightarrow Z$ - statistic is the appropriate one

- **Step 4: $\alpha = 0.05$**
- **Step 5:**

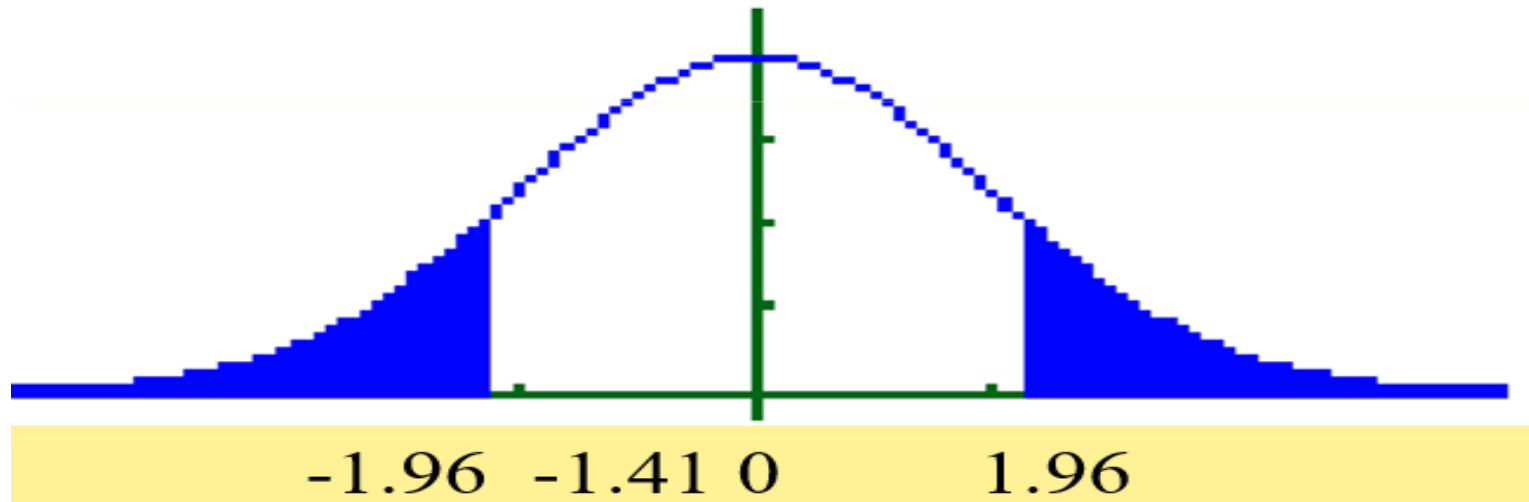


- **Step 6:**

- **Given**

- **Sample Mean = 22, $\mu_0 = 25$, $\sigma^2 = 45$, $n = 10$**

$$\Rightarrow Z = \frac{22 - 25}{\sqrt{45 / 10}} = -1.41$$



- **Step 7:**

- **Since -**

- 1.41 falls in the acceptance region we accept the null hypothesis.**

- **The mean enzyme level in the population is not different from 25.**

- **Unknown variance**
 - **The test statistic is t - test**

$$t = \frac{\bar{X} - \mu_0}{S / \sqrt{n}}$$

- **Example**

- **Serum amylase determination were made on a sample of 15 apparently health subjects.**
- **The sample yielded a mean of 96 units/100ml and a standard deviation of 35 units/100ml. The population was normally distributed but the variance was unknown.**
- **We want to know whether we can conclude that the mean of the population is different from 120.**

- **Solution**

- **Step 1:**

- **H0: $\mu = 120$**
- **HA: $\mu \neq 120$**

- **Step 2:**

- **They collect a sample of size 15 from a normally distributed population with a mean = 96, SD = 35, n =15**

- **Step 3:**

- **Testing a hypothesis about population mean**
- **The population is normally distribute**
- **Population variance is unknown**
- **$\Rightarrow$ t -statistic is the appropriate one**

$$t = \frac{\bar{X} - \mu_0}{S / \sqrt{n}}$$

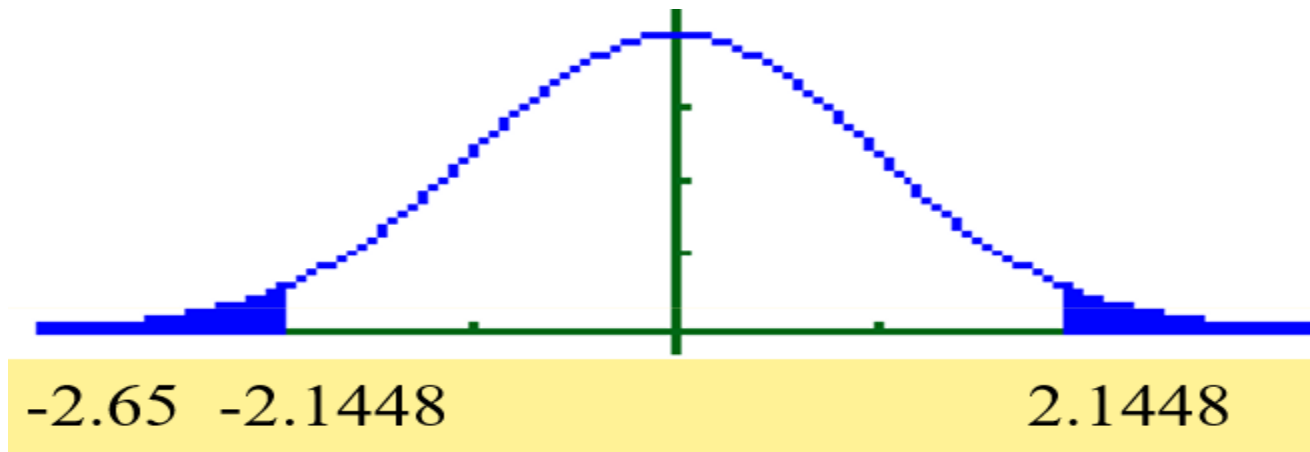
- **Step 4:**

- $\alpha = 0.05$, $t_{0.025, 14} = 2.1448$

- **Step 5:**

$$t = \frac{96 - 120}{35 / \sqrt{15}} = -2.65$$

- Since $-2.65 < -2.1448$ it falls in the rejection region we reject the null hypothesis.
 - The mean of the population from which the sample came is not 120.



- **Exercise**

- **A simple random sample of 100 people from a certain population has a mean age of 27. Can we conclude that the mean age of the population is not 30? The population variance is known to be 20. Let $\alpha = 0.05$.**
- **A simple random sample of 100 people from a certain population has a mean age of 27. Can we conclude that the mean age of the population is less than 30? The population variance is known to be 20. Let $\alpha = 0.05$.**

- **Hypothesis testing about a single population proportion**
- **Example:**
 - **A survey was conducted to study the dental health practices, and attitudes of a certain urban adult population. Of 300 adults interviewed, 123 said that they regularly had a dental check up twice a year. Can we conclude that the population proportion $\pi = 0.5$?**

Solution

$$H_0 : \pi = 0.5$$

$$H_1 : \pi \neq 0.5$$

$$P = 123/300 = 0.41$$

$$\text{Test statistic, } Z = \frac{p - \pi_0}{\sqrt{\frac{\pi_0(1-\pi_0)}{n}}}$$

$$\alpha = 0.05, Z_{0.025} = 1.96$$

$$Z = \frac{0.41 - 0.5}{\sqrt{\frac{0.5(1-0.5)}{300}}} = -3.11$$

Since $-3.11 < -1.96$ we reject H_0 .

We conclude that not 50% of the population regularly have a dental check up twice a year.

- **The Chi – square test**
 - **A chi square is useful in making statistical inferences about categorical data in which the categories are two and above .**
 - **Definition: A statistic which measures the discrepancy between observed frequencies and the corresponding expected frequencies**

$$\text{Chi square} = \chi^2 = \sum \{ (O_i - e_i)^2 \} / e_i$$

Characteristics Chi - square

- Every χ^2 distribution extends indefinitely to the right from 0.
- Every χ^2 distribution has only one (right) tail.
- As df increases, the χ^2 curves get more bell shaped and approach the normal curve in appearance (but remember that a chi square curve starts at 0, not at negative infinity).
- If the value of χ^2 is zero, then there is a perfect agreement between the observed and the expected frequencies.
- The greater the discrepancy between the observed and expected frequencies, the larger will be the value of χ^2

Steps in the Analysis

Step 1. State the hypotheses

- The null hypothesis (H_0) states no association between exposure and disease
- The alternative hypothesis (H_A) states there is an association between exposure and disease

Step 2. Data

- Construct a 2x2 table with cells comprised of observed data or sample frequencies
- Calculate the expected frequencies

Step 3. Test statistics

- Chi-square test with $df=1$

- **Step 4.** Calculate the χ^2 value
- **Step 5.** Reject H_0 if the computed value of chi-square is $\geq$ the tabulated chi-square for the chosen value of α
- **Step 6.** Give interpretation and conclusion

Example

- A study was conducted to look at the effects of oral contraceptives (OC) on heart disease (MI) in women 40 to 44 years of age. It is found that among 5000 current OC users at baseline, 13 women developed an MI over a 3-year period, where as among 10,000 non-OC users, 7 developed an MI over a 3-year period (**Prospective Study**).

- **Solution:**
- **Step 1:** state the hypothesis
 - **Ho:** MI is the same in OC users and no OC users.
 - **HA:** MI is different in OC users and non OC users.
- **Step 2:** construct 2x2 table

OC use group	MI Status over 3 years period		Total
	Yes	No	
OC- Users	13	4987	5000
OC- Non users	7	9993	10,000
Total	20	14,980	15,000

- Step 3: test statistics with df:
- **Degrees of Freedom (df)**
 - Chi-Square test is based on the table of χ^2 for different degrees of freedom (df).
 - χ^2 degree of freedom is given by:
 $(\text{no. of rows}-1) * (\text{no. of cols}-1)$
 $= (R-1)(C-1)$

- The general calculation:

$$\chi^2 = \sum_{\text{all cells}} \frac{(O_{ij} - E_{ij})^2}{E_{ij}}$$

Calculation of Expected Values

- The expected frequency in each cell is the product of the row and column totals divided by the sum of all the observed frequencies (i.e., sample size)

$$E = \frac{\text{row total} \times \text{column total}}{\text{Overall total}}$$

- The expected numbers must be computed for each cell.
- The sum of the corresponding observed and expected row and column totals should be the same.

- Compute the **expected table** for the OC-MI data

OC use group	MI status over 3-years		Total
	Yes	No	
OC users	6.7	4993.3	5000
Non-OC users	13.3	9986.7	10,000
Total	20	14,980	15,000

- Calculate $\chi^2 = 9$
- Read χ^2 - tab with $df = 1$ and $\alpha = 0.05$
- χ^2 - tab = 3.841

- **Step 4: set α (usually= 0.05)**
- **Step 5: reject H_0 if $\chi^2 - \text{cal} \geq \chi^2 - \text{tab}$**
 - **Therefore $9 > 3.841$ H_0 is rejected**
- **Step 6: give interpretation and conclusion**
 - **There is an association between taking OC and MI.**

Quiz 1

- A study was conducted to investigate the possible cause of gastroenteritis outbreak following a lunch served in a high school cafeteria. Among the 225 students who ate the sandwiches, 109 became ill. While, among the 38 students who did not eat the sandwiches, 4 became ill (Outbreak investigation, cross-sectional).

Estimation

- **Estimation**

- **Statistics can be broadly classified as descriptive and inferential**
- **Inferential statistics also divided in to estimation and hypothesis testing**
- **Estimation – Describing population parameter in terms of sample statistics.**
- **Parameter – is a numerical descriptive measures of a population data.**

- **Statistics:** is a numerical descriptive measures of sample data.
- A sample from population is used to provide the **estimate of the population parameter.**
- To each sample statistics we use corresponding population parameters.

- Cont...

Sample statistic

$\bar{X}$ (Sample mean)

S (Sample SD)

S^2 (Sample Variance)

p (Sample Proportion)

Corresponding
population parameter

μ (Population mean)

σ (Population SD)

σ^2 (Population Variance)

P or π (Population
proportion)

- **Types of Estimation**

- **Point Estimation**

A single numerical value used to estimate the corresponding population parameter

- $\bar{x}$ is an estimator of the population mean μ
 - S is an estimator of the population standard deviation σ
 - P is an estimator of the population proportion π
 - r is an estimator of the population correlation coefficient ρ
 - $OR^{\hat{}}$ is an estimator of population OR
 - $RR^{\hat{}}$ is an estimator of population RR

- **Example:** the plasma volume/ liter of a sample of eight healthy adult male were as follows.
- **2.75, 2.86, 3.37, 2.76, 2.62, 3.49, 3.03, 3.12, liters**
- **Estimate the mean and SD of plasma volume for a total population of healthy adult male.**

- **Interval estimation**
 - **Point estimation does not give any indication on how far away the parameter lies.**
 - **A more useful method of estimation is to compute an interval which has a high probability of containing the parameter.**
 - **Interval estimation** - is an estimate of parameter in terms of an interval or range of value with which it is likely to lie.

• Cont...

- **Interval estimation is better than point estimation because it considers the variation within the samples.**
- **The interval with in which population parameter is assumed to lie is confidence interval.**
- **The upper and the lower values of the interval are called confidence limit.**
- **The population parameter lies in interval and it is not 100% sure but we have some confidence that the population parameter may lie on this interval.**

- **Cont...**

- **Two questions to put bounds on our point estimates to reflect our level of confidence.**

- **How wide does the bracket have to be?**

- **What is our tolerance of error (variability)?**

- **Scientists usually accept a 5% chance that the range will not include the true population value**

- **The range or interval is called 95% confidence interval**

100(1- α)% CI for μ when σ is known (sampling from normal population or large sample)

estimator

precision of the estimate (margin of error)

$$\bar{x} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$

reliability coefficient

standard error

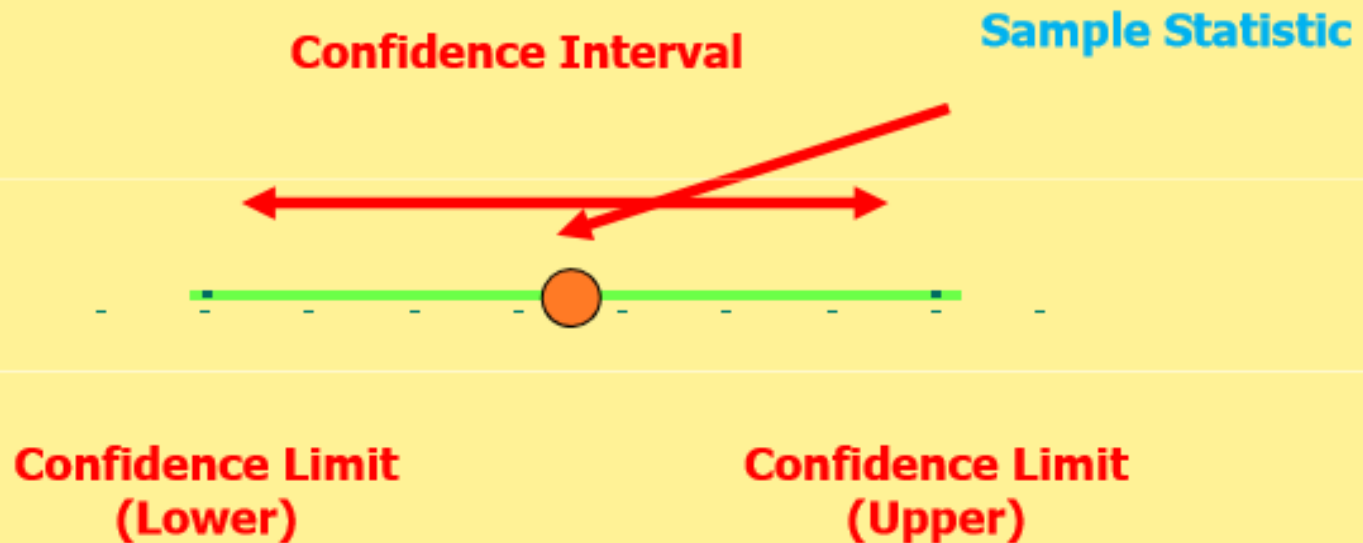
The diagram shows the formula for a confidence interval: $\bar{x} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$. Arrows point from descriptive labels to parts of the formula: 'estimator' points to $\bar{x}$; 'precision of the estimate (margin of error)' points to the entire $\pm$ term; 'reliability coefficient' points to $z_{\alpha/2}$; and 'standard error' points to $\frac{\sigma}{\sqrt{n}}$.

Interpretation:

- Probabilistic: in repeated sampling, 100(1- α)% of all intervals will include μ
- Practical: we are 100(1- α)% confident that a single interval contains μ

Interval estimation

A Probability That the Population Parameter Falls Somewhere Within the Interval.



- **Factors Affecting Interval Width**
 - **Data Variation**
 - **Sample Size**
 - **Level of Confidence**

Intervals Extend from

$$\mu - Z_{\alpha/2} \sigma_{\bar{X}}$$

to

$$\mu + Z_{\alpha/2} \sigma_{\bar{X}}$$

- **C.I. for a population mean
(normally distributed)**

- **Known variance (large sample size)**

- **A 100(1-α)% C.I. for μ is**

$$\bar{x} \pm Z_{\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}} \Rightarrow \left(\bar{x} - Z_{\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}}, \bar{x} + Z_{\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}} \right)$$

- **α is to be chosen by the researcher, most common values of α are 0.05, 0.01, 0.001 and 0.1**

- **Interpretation:**

- **Practical:** we are 100(1-α)% confident that an interval contains μ

- **Example**

- **A data on 199 patients on systolic blood pressure gives a mean value of 125.8mmHg. Let us assume that the standard deviation for this patient population is known to be 20 mmHg.**
- **Construct a 95% confidence interval for the population mean.**

- **Solution**

- $\alpha = 0.05 \Rightarrow Z_{\alpha/2} = 1.96$

- $125.8 \pm 1.96 \times 20/14.107$

- The 95% CI is (123.0, 128.6 mmHg)

- We are 95% sure that the average SBP
for patients population is
between 123 and 128.6.

- **Unknown variance
(small sample size $n \leq 30$)**

- **A 100 (1- α)% C.I. for μ is**
- **We use here Students' t-distribution for determination of confidence interval**

$$\bar{X} \pm t_{\frac{\alpha}{2}, n-1} \frac{S}{\sqrt{n}} \Rightarrow \left(\bar{X} - t_{\frac{\alpha}{2}, n-1} \frac{S}{\sqrt{n}}, \bar{X} + t_{\frac{\alpha}{2}, n-1} \frac{S}{\sqrt{n}} \right)$$

- **The t-distribution density curve is bell shaped and symmetrical about zero.**
- **Different curves for different df (i.e sample sizes) and for very large df very close to Z**

- **Example**

- **In a study of preeclampsia, Kaminski and Rechberger found the mean systolic blood pressure of 10 healthy, non-pregnant women to be 119 with a standard deviation of 2.1.**

- What is the estimated standard error of the mean?**
 - Construct the 99% confidence interval for the mean of the population from which the 10 subjects may be presumed to be a random sample.**
 - What is the precision of the estimate?**

• Solution

A. $SE = \frac{S}{\sqrt{n}} = \frac{2.1}{\sqrt{10}} = 0.66$

B. $df = n - 1 = 10 - 1 = 9$

99%  $\alpha = 0.01$  $\alpha/2 = 0.005$

$t_{0.005,9} = 3.250$

$$\bar{X} \pm t_{\alpha/2, n-1} \frac{S}{\sqrt{n}} \Rightarrow 119 \pm 3.250 \times 0.66$$

The 99% CI is (116.8, 121.2)

C. Precision = 3.250×0.66
 $= 2.16$

- **C.I. for a population proportion (large sample size)**
 - **A 100(1- α)% C.I. for π is:**

$$p \pm Z_{\frac{\alpha}{2}} \sqrt{\frac{p(1 - p)}{n}}$$

- **Example**

- **A research study obtained data regarding sexual behavior from a sample of unmarried men and women between the ages of 20 and 44 residing in geographic areas characterized by high rates of sexually transmitted diseases and admission to drug programs.**
- **Fifty percent of 1229 respondents reported that they never used a condom.**
- **Construct a 95 percent confidence interval for the population proportion never using a condom.**

- **Solution**

Given

$$n = 1229$$

$P = .50$ a point estimator of π

$$\alpha = 0.05 \Rightarrow Z_{0.025} = 1.96$$

$$0.5 \pm 1.96 \sqrt{\frac{0.5 \times 0.5}{1229}}$$

The 95% CI is (0.47, 0.53)

Sample Size Estimation

- **What is sample size?**

- **Sample size:-** the number of study population required to estimate a population parameters

- **Researchers always ask themselves “How big sample do I need?”**

- **Too large sample size: too expensive and time consuming**

- **Too small sample size: it has inadequate precision to show a good estimate or show difference**

- **Sample size**

- **Sample size is dependent on the type of design,**
 - **Descriptive vs analytic**
 - **Case control vs Cohort**
 - **Analytic cross sectional vs Case control**
- **It is also dependent on the type of major variable used (categorical vs continuous variable)**

- **Steps in Estimating Sample Size**

- **Identify major study variable**
- **Determine type of estimate (percentage, mean, ratio,...)**
- **Indicate expected frequency of factor of interest**
- **Decide on desired precision of the estimate**

- **Cont... on steps**
- **Decide on acceptable risk that estimate will fall outside its real population value**
- **Adjust for population size**
- **Adjust for estimated design effect**
- **Adjust for expected response rate**

- **Sample size depends on**
 - **Estimated variability**
 - **The precision (margin of error)**
 - **The sampling method (clustering, design effect)**
 - **Size of population**
 - **Feasibility (cost)**
 - **Confidence level (Z value of certainty)**

- **Why is it important to consider sample size?**

- **In studies concerned with estimating some characteristic of a population (e.g. the prevalence of asthmatic children),**
- **Sample size calculations are important to ensure that estimates are obtained with required precision or confidence.**

- **For example**

- **A prevalence of 10% from a sample of size 20 would have a 95% confidence interval of 1% to 31%, which is not very precise or informative.**
- **On the other hand, a prevalence of 10% from a sample of size 400 would have a 95% confidence interval of 7% to 13%, which may be **considered sufficiently accurate**.**
- **Sample size calculations help to avoid this situation.**

- Therefore, to obtain the **optimum sample size**, decision on the following is important.
 - How large **error** can be tolerated during estimation (d)
 - **Confidence limit** that the tolerated error will not exceed the determined one ($1 - \alpha$)
 - Our advanced guess of **population variance** or **proportion**.

- **Where do we get this knowledge?**
 - **Previous published studies**
 - **Pilot studies**
 - **If information is lacking, there is no good way to calculate the sample size.**

- The **absolute precision** (d) is half of the confidence interval.

$$d = Z_{\frac{\alpha}{2}} SE$$

where SE is the standard error of the estimator of the parameter of interest

- Estimating a single Population Mean

$$d = Z_{\frac{\alpha}{2}} SE$$

$$= Z_{\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}}$$

$$\Rightarrow \sqrt{n} = Z_{\frac{\alpha}{2}} \frac{\sigma}{d}$$

$$\Rightarrow n = Z_{\frac{\alpha}{2}}^2 \frac{\sigma^2}{d^2}$$

- **To estimate a single population proportion**

$$d = Z_{\frac{\alpha}{2}} SE$$

$$= Z_{\frac{\alpha}{2}} \sqrt{\frac{P(1-P)}{n}}$$

$$\Rightarrow \sqrt{n} = Z_{\frac{\alpha}{2}} \frac{\sqrt{P(1-P)}}{d}$$

$$\Rightarrow n = Z_{\frac{\alpha}{2}}^2 \frac{P(1-P)}{d^2}$$

- **Failure to Achieve Required Sample Size**

- **Patient refusal to consent**
- **Bad time of the study (heavy working time for participants)**
- **Adverse media publicity**
- **Weak recruiting staff**
- **Lack of genuine commitment to the project**
- **Lack of staffing in wards or units**
- **Too many projects attempting to recruit the same subjects**

- **Possible Solutions**

- **Pilot studies**
- **Have a plan to regularly monitor recruitment or create recruitment targets**
- **Ask for extension in time and/or funding**
- **Review your staffs commitment to other ongoing trials or other distracters**
- **Regular visits to field sites**

THANK YOU